7. The price of three commodities X, Y and Z are x, y and z respectively Mr.Anand purchases 6 units of Z and sells 2 units of X and 3 units of Y. Mr.Amar purchases a unit of Y and sells 3 units of X and 2units of Z. Mr.Amit purchases a unit of X and sells 3 units of Y and a unit of Z. In the process they earn ₹5,000/-, ₹2,000/- and ₹5,500/- respectively Find the prices per unit of three commodities by rank method.

8. An amount of ₹5,000/- is to be deposited in three different bonds bearing 6%, 7% and 8% per year respectively. Total annual income is ₹358/-. If the income from first two investments is ₹70/- more than the income from the third, then find the amount of investment in each bond by rank method.

1.2 Cramer's Rule

Gabriel Cramer, a Swiss mathematician born in the city Geneva in 31 July 1704. He edited the works of the two elder Bernoulli's, and wrote on the physical cause of the spheriodal shape of the planets and the motion of their apsides (1730), and on Newton's treatment of cubic curves (1746).

In 1750 he published Cramer's Rule, giving a general formula for the solution of certain linear system of n equations in n unknowns having a unique solution in terms of determinants. Advantages of Cramer's rule is that we can find the value of x, y or z without knowing any of the other values of x, y or z. Cramer's rule is applicable only when $\Delta \neq 0$ (Δ is the determinant value of the coefficient matrix) for unique solution.

Theorem (without proof) Cramer's Rule:

Consider, $\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$

If $AX = B$ is a system of n linear equations in 'n' unknowns such that $\det(A) \neq 0$, then the system has a unique solution.

This solution is,

$$x_1 = \frac{\det(A_1)}{\det A}, \quad x_2 = \frac{\det(A_2)}{\det A}, \ \dots \ , \ x_n = \frac{\det(A_n)}{\det A}$$

where A_j is the matrix obtained by replacing the entries in the jth column of A by the entries in the matrix $B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$

1.2.1 Non Homogeneous linear equations upto three variables

(a) Consider the system of two linear equations with two unknowns.

$$a_1x + b_1y = d_1$$

$$a_2x + b_2y = d_2$$

Let $\Delta = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$ $\Delta_x = \begin{vmatrix} d_1 & b_1 \\ d_2 & b_2 \end{vmatrix}$ $\Delta_y = \begin{vmatrix} a_1 & d_1 \\ a_2 & d_2 \end{vmatrix}$

The solution of unknown by Cramer's rule is

$$x = \frac{\Delta x}{\Delta}, \qquad y = \frac{\Delta y}{\Delta} \qquad \text{provided } \Delta \neq 0$$

(b) Consider the system of three linear equations with three unknowns

$$a_1x + b_1y + c_1z = d_1$$
$$a_2x + b_2y + c_2z = d_2$$
$$a_3x + b_3y + c_3z = d_3$$

Let $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \neq 0$ $\Delta_x = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$

$\Delta_y = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$ $\Delta_z = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$

Solution of unknown by Cramer Rule is

$$x = \frac{\Delta x}{\Delta}, \ y = \frac{\Delta y}{\Delta}, z = \frac{\Delta z}{\Delta}$$

XII Std - Business Maths & Stat EM Chapter 1.indd 14 2/28/2020 5:28:02 PM